

Madhur Lakshmanan

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SUMMARY

Applied AI Engineer with an MS in Applied Machine Learning (University of Maryland, GPA 3.8) who ships LLM applications end to end: RAG, multi-agent orchestration with tool calling, schema-constrained extraction, and the APIs and cloud infrastructure behind them. Built LLM platform components at Causify AI.

EXPERIENCE

Causify AI | *Machine Learning Engineer Intern*

July 2025 – September 2025

- Built core components of an LLM-orchestrated AutoEDA platform on FastAPI and self-hosted Ollama models, implementing schema profiling, structured outputs, and agent-tool interfaces so users could analyze datasets in natural language
- Designed scikit-learn trend, changepoint, outlier, and association detection tools as executable interfaces for the LLM-driven workflow
- Built an internal documentation assistant with Dify, automating Markdown ingestion and knowledge-base sync via GitHub Actions

Thomson Reuters | *AI Technology Intern*

March 2024 – August 2024

- Re-engineered a legacy OmniMark workflow into parallelized Python and Bash pipelines integrating REST APIs and Microsoft SQL Server, cutting processing time by nearly 70%
- Developed time-series forecasting pipelines in AWS SageMaker, improving forecast accuracy by 18% over the production baseline

Mareana | *Data Intern*

June 2022 – July 2022

- Documented REST APIs for 100+ data endpoints, reducing developer onboarding time by 40% and supporting downstream analytics and ML pipeline integration

PROJECTS

PersonalOS: Multi-Agent Job-Search Assistant | *LangGraph, MCP, FastAPI, PostgreSQL, pgvector, Alembic*

[GitHub](#)

- Building a local-first agentic assistant in LangGraph: a supervisor graph with structured-output routing routes to a job-search subgraph that deduplicates, scores, ranks, and shortlists postings
- Designed a default-deny security model in which the LLM emits only tool intents; a policy engine allows, gates for human approval, or denies each one, and an isolated executor runs approved actions exactly once per idempotency key with a full audit trail
- Checkpointed workflow state to PostgreSQL for crash-safe resume, exposed tools through MCP servers and a FastAPI service, and covered the system with 600+ tests and CI-enforced layer boundaries

Serverless Research Intelligence Platform | *Bedrock, Titan embeddings, pgvector, Lambda, AWS CDK*

[GitHub](#)

- Built a serverless RAG system that ingests arXiv papers, embeds them with Amazon Titan (1024-dim), and serves semantic search and chat over Aurora PostgreSQL/pgvector
- Returned retrieved passages, paper metadata, and similarity scores with every answer for citation-grounded output

Financial & Regulatory Document Intelligence | *OpenAI GPT-4, LangChain, SEC EDGAR API, MongoDB*

[GitHub](#)

- Designed schema-constrained GPT-4 extraction with few-shot prompting and validated JSON schemas for risk factors, litigation, financial metrics, and material events across 500+ companies and 15+ years of filings
- Orchestrated multi-stage LLM workflows for extraction, validation, retrieval, summarization, and report generation with auditable structured outputs

EDUCATION

University of Maryland, College Park | *M.S., Applied Machine Learning* (GPA: 3.8/4.0)

May 2026

Vellore Institute of Technology | *B.Tech., Electronics and Communication Engineering* (GPA: 3.2/4.0)

May 2024

SKILLS

LLM & Agents: LangChain, LangGraph, MCP, RAG, pgvector, Amazon Bedrock, OpenAI API, Ollama, vLLM, Dify, structured outputs, tool calling, prompt engineering

Backend & Cloud: FastAPI, PostgreSQL, Redis, Celery, SQLAlchemy, AWS (Lambda, Bedrock, CDK, SageMaker, S3), Docker, Kubernetes, GitHub Actions

Languages & ML: Python, SQL, Java, C/C++, Bash, PyTorch, Hugging Face, scikit-learn